EE6301
Study Priorities
重点复习路线

First pass · Foundations · Supporting reading
P1先掌握 · P2重要基础 · P3辅助阅读

Body text / 正文：11 pt
English first, Chinese immediately below. / 英文在前，中文紧随。

2026-10-13 · Course notes, sample explanations and original practice
课件知识、样题参考解析与原创练习

Contents / 内容目录

Study priorities and route / 重点标记与复习路线 3

Study priorities and route / 重点标记与复习路线

EN: P1: master first and explain each statement; P2: important foundations that connect or qualify P1; P3: supporting details to consult after the first pass. These levels are study recommendations, not probabilities or a guarantee of the real questions.

中文： P1：先掌握，能解释陈述；P2：重要基础，用来连接P1及理解条件；P3：辅助细节，第一轮后按需查。等级是复习建议，不是出题概率或真题保证。

EN: Evidence tags are separate: Quiz stated, teacher emphasis, repeated explanation, worked example, sample quiz and final-only emphasis. “Important” alone is not a Quiz promise. Example tags identify practice in the slides/class; they do not invent a separate homework assignment.

中文： 依据标签分别为：Quiz明确提及、课堂强调、反复讲解、例题练习、样卷涉及、仅期末强调。“重要”本身不等于Quiz保证。例题标签指课件或课堂练习，不凭空认定为独立课后作业。

EN: A mixed lesson has a specific review target: focus on that part first. A sample question marks all its relevant option concepts. Its extra derivations or device details can still be P2/P3. Open “Evidence” below a heading in the HTML to see the review target, transcript references and the lecturer’s short quote when available.

中文： 混合小节有具体学习抓手，先学该部分；样卷标记覆盖相关选项的概念，额外推导或器件细节仍可为P2/P3。HTML中展开标题下的“依据”，可看学习抓手、字幕位置及可核对的老师短引语。

P1: 44 • P2: 53 • P3: 12

First-pass study group / 第一轮学习组	P1 locations / P1位置	What to do / 要会什么
Measurement and LOD / 测量性能与检测限	C02, C03, B03	Distinguish slope, selectivity and LOD; use the linear range and integer-count rule. / 区分斜率、选择性与LOD；会选线性区、计算及整数计数处理。
Photon energy and absorption / 光子与吸收	C04, S105	Match wavelength, energy and photon count; calculate absorbance, transmission and ε. / 对应波长、能量、光子数；会算吸光度、透过率与ε。
Fluorescence / 荧光	C05, C06, S110, S111	Explain intensity versus color, source/filter roles and Φ, τ, radiative rate. / 解释强度与颜色、光源及滤光片作用；区分Φ、τ与辐射速率。
Raman and scattering / 拉曼与散射	C07, B08	Distinguish elastic/inelastic signals and Stokes/anti-Stokes; read cm^{-1}. / 区分弹性/非弹性与Stokes/anti-Stokes；会读cm^{-1}及位移。

First-pass study group / 第一轮学习组	P1 locations / P1位置	What to do / 要会什么
Grating resolution / 光栅分辨	C08	Use R=mN and distinguish groove density, illuminated width and diffraction order. / 会用R=mN，区分刻线密度、照明宽度与衍射级次。
Label-free, fibre and resonance / 无标记、光纤与共振	C09, C10, C11, B10, S115, S116, S117	Recognize cavities, SPR and Raman; explain TIR, evanescent sensing and spatial arrangements. / 辨认共振腔、SPR与Raman；解释全反射、倏逝场及光纤传感布置。
ART and LSPR / 抗原快检与LSPR	C12	Separate antibody recognition from gold-particle color generation and check the control line. / 区分抗体识别与金颗粒显色，并会判断控制线。
Microscopy and resolution / 显微成像与分辨	C13, C14, B14, B15, B17	Compare NA rather than magnification alone; distinguish confocal and two-photon mechanisms. / 比较NA而非只看倍率；区分共聚焦与双光子机制。
OCT and light detectors / OCT与光探测器	C15, C16, S404	Explain reflectivity-based OCT and CCD/CMOS/PMT roles; separate depth and lateral resolution. / 理解OCT反射读出与CCD/CMOS/PMT；分清轴向、横向分辨。
Ultrasound and photoacoustics / 超声与光声	B18, S501	Trace transmit/receive and piezoelectric conversion; distinguish echo depth from a photoacoustic source path. / 会追踪发射接收与压电转换；区分回波深度与光声源传播路径。
Electrochemical readouts / 电化学读出	C18, C19, S603	Match voltage/current/impedance readouts, electrode roles and GOx reaction-current relations. / 对应电位/电流/阻抗模式、电极分工与GOx反应—电流关系。
Bioelectricity / 生物电基础	C20, C21, B27, S707	Know sources of biopotentials, resting/spike phases and basic charge/voltage/current units. / 理解生物电来源、静息及尖峰阶段，记基础电荷、电压、电流单位。
ECG, EEG and watches / 心电、脑电与手表	C22, S705, S706	Identify wave sources and time intervals; estimate frequency and separate watch PPG from EEG/EMG. / 辨认波形来源与时间；估频率，区分手表PPG与EEG/EMG。

First-pass study group / 第一轮学习组	P1 locations / P1位置	What to do / 要会什么
Patch clamp and MEA / 膜片钳与多电极阵列	C23, S801	Recognize pipette versus array, intracellular/extracellular signals and throughput differences. / 辨认吸管与阵列、胞内与胞外，以及通量差别。
Amplification and filters / 放大与滤波	C24, C25	Separate gain, input/output impedance and common-mode rejection; recognize four filter curves. / 区分增益、输入输出阻抗与共模抑制；会认四种滤波曲线。

EN: Suggested order: read the prerequisites you need, complete the P1 groups, use P2 to explain unfamiliar options, then take the 25-question mock in English. Return to P3 when a concept or a diagram still blocks understanding. Keep the full guide for checking; a P3 label does not mean a whole field is excluded.

中文： 建议顺序：先补自己缺的先修基础，完成P1学习组；用P2解释不熟悉的选项，再用纯英文做25题。概念或图仍不懂时回查P3。完整资料仍供查阅；P3不等于整个领域已被排除。